Linux Fundamentals
Project: Info Extractor

System Diagnostics Report

1. Identify the system's public IP

The public IP address is the external IP assigned by the Internet Service Provider and is used to communicate with systems outside the local network.

```
curl -s ifconfig.me
```

- curl: Tool for transferring data from or to a server
- -s: Silent mode
- ifconfig.me: An online service that returns your public IP

2. Identify the private IP address assigned to the system's network interface

The private IP is assigned to the system by the router within a local network. It is used for internal communication.

```
ifconfig | grep broadcast | awk '{print $2}'
```

- ifconfig: Displays network interface configurations
- grep broadcast: Filters the line that contains the IP address
- awk '{print $2}': Extracts the second column (the IP)

3. Display the MAC address (masking sensitive portions for security)

The MAC address is a unique hardware identifier assigned to the system's network card. For privacy reasons, part of it is masked.

```
FULL_MAC_ADDRESS=$(ifconfig | grep ether | awk '{print $2}')
MASKED_MAC="${FULL_MAC_ADDRESS:0:8}:XX:XX:XX"
```

- The MAC address has 6 pairs of hexadecimal digits, separated by colons

- ${FULL_MAC_ADDRESS:0:8}: extracts the first 3 pairs
- :XX:XX:XX is appended to hide the unique part for privacy

4. Display the percentage of CPU usage for the top 5 processes

Monitoring CPU usage helps identify which processes are resource-intensive.

```
ps -eo %cpu,comm --sort=-%cpu | head -n 6
```

- ps -eo %cpu,comm: Lists all processes showing only CPU usage and command
- --sort=-%cpu: Sorts processes by CPU usage in descending order
- head -n 6: Displays the top 5 processes plus header line

5. Display memory usage statistics: total and available memory

Understanding memory usage helps diagnose performance issues.

```
free -h | awk '/Mem:/ {print "Total: "$2", Available: "$7}'
```

- free -h: Displays memory info in human-readable format
- /Mem:/: Filters only the line showing RAM
- $2, $7: Show total and available memory

6. List active system services with their status

Checking active services is key for understanding what's running in the background.

```
service --status-all | grep -F '[ + ]'
```

- service --status-all: Lists all system services and their statuses
- grep -F '[+]': Filters only those that are currently running

7. Locate the Top 10 Largest Files in /home

Identifying large files helps optimize disk space usage and manage storage efficiently.

```
find /home -type f -exec du -b {} + | sort -nr | head -n 10
```

- find /home -type f: Find all files under /home
- du -b: Prints the file size in bytes, for precise sorting
- sort -nr: Sorts files by size in descending order
- head -n 10: Displays the ten largest files

Conclusion

This report provides a quick but powerful set of tools to diagnose key aspects of a Linux system, including networking, memory, CPU, services, and disk usage. Each section offers practical shell commands with explanations suitable for real-world use in system administration and troubleshooting.

Tools and Commands used: curl, ifconfig, grep, awk, ps, head, free, service –status-all, find, du, sort